Pakistan Navy Complaint Management System (CMS)

Document 07: Comprehensive System User & Administration Manual

Document Identifier: CMS-OPS-V1.2.4
Target Audience: Helpdesk Operators, Garrison Engineers (GE / Supervisors), CMES / XEN Officers, Storekeepers, System Administrators
Classification: Restricted (Internal Pakistan Navy Housing & Engineering Directorate)
Date: September 2026
Status: Approved for Operations & Handover

1. Document Control & Operational Overview

1.1 Purpose

This manual serves as the standard operational guide and administrative reference for the **Pakistan Navy Complaint Management System (CMS)**. It specifies exact operating procedures for handling residential complaints, coordinating engineering dispatches, sanctioning material requisitions through the Performa engine, managing warehouse stock, monitoring SLA compliance, and administering user roles.

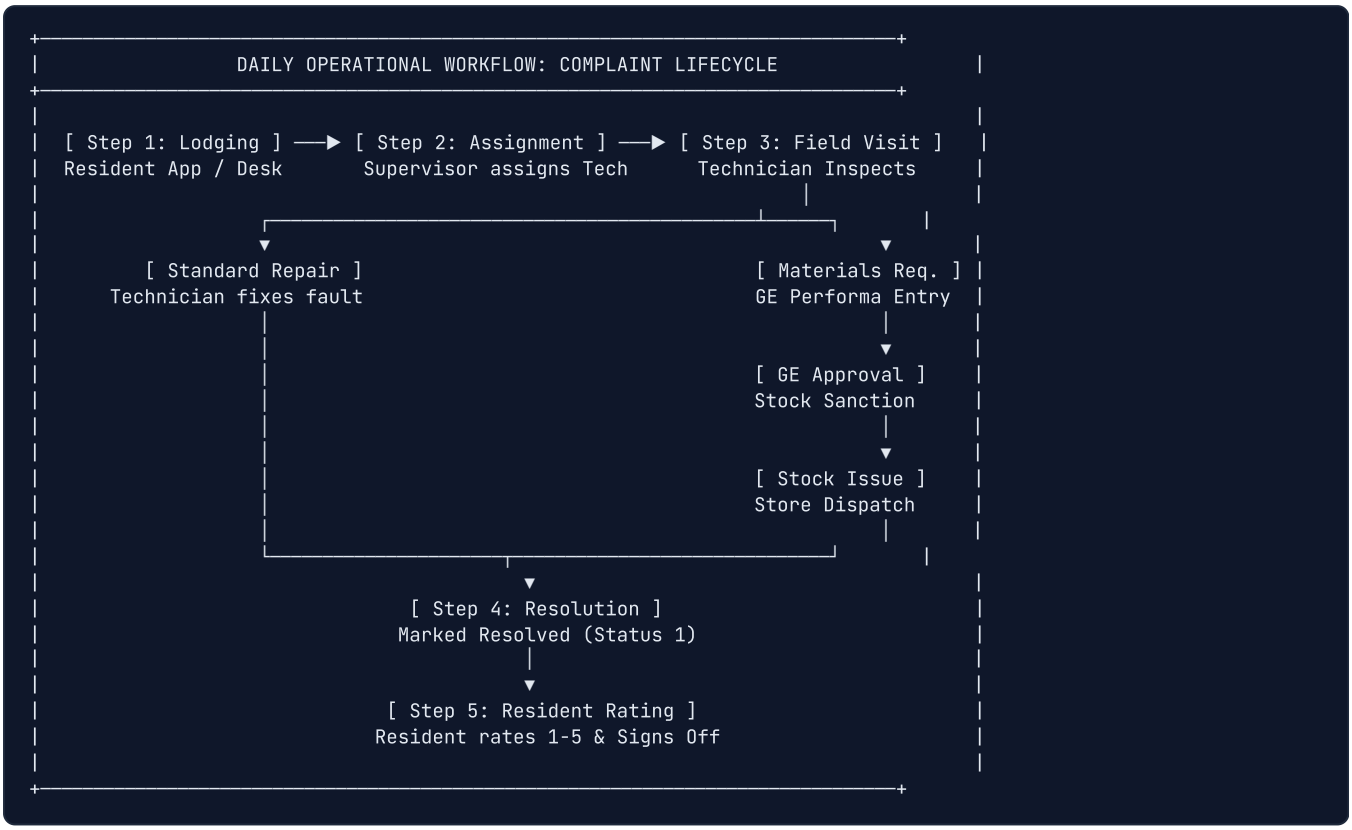
1.2 Target Stakeholder Personas

The system provides tailored interfaces and workflows based on six core operational roles:

Role Title	Operational Responsibilities	Primary Interface Modules
Super Administrator	Platform governance, RBAC permissions, audit logging, master catalog configuration	All Modules (`/admin/*`)
CMES / XEN Officer	Strategic oversight, financial sanction of priced performas, SLA compliance audit	Executive Dashboard, Approvals, Reports
Garrison Engineer (GE) / Supervisor	Ticket triage, technician allocation, work order generation, technical inspection	Complaints Board, Assign, Performas, Staff
Helpdesk / Control Room Operator	Call/walk-in intake, resident data validation, ticket registration, status follow-up	Register Complaint, Board, Houses, Devices
Storekeeper / Inventory Manager	Inward purchase entries, outward material issuance against vouchers, stock ledger	Spares Catalog, Stock Inward, Issue Slips
Resident / Naval House Occupant	Submitting maintenance requests, selecting availability slots, rating work	Mobile App / Resident Web Portal

2. Daily Operational Workflow: End-to-End Complaint Lifecycle

The following workflow illustrates the end-to-end operational path of a maintenance request from initial registration to financial reconciliation and resident sign-off:



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3. Helpdesk Operations: Complaint Intake & Ticket Registration

The Helpdesk / Control Room is the front-line interface for capturing resident issues via phone call, radio transmission, or walk-in service desks.

3.1 Step-by-Step Ticket Registration

1. Navigate to **Complaints** → **Register Complaint** (`/admin/complaints/create`).
2. **Geographic Identification (Cascading Filters):**
   * **City:** Select station (e.g. *Karachi*, *Islamabad*).
   * **CMES Formation:** Select the responsible engineering unit (e.g. *CMES (Navy) Karachi*).
   * **Housing Sector:** Select the designated colony/zone (e.g. *Karsaz*, *Naval Heights*).
   * **House / Accommodation:** Type or select the accommodation number from the predictive search.
3. **Resident Verification:**
   * System automatically displays the registered occupant's Rank, Name, Official ID, and Contact Number.
   * If details differ from caller, verify authorization before proceeding.
4. **Issue Classification:**
   * **Category:** Choose technical discipline (*Electrical*, *Civil / Plumbing*, *Carpentry*, *Sui Gas*, *HVAC*).
   * **Sub-Category:** Select defect group (e.g. *Ceiling Fan*, *Internal Water Supply*, *Main Circuit Breaker*).
   * **Complaint Title:** Select standard fault title (e.g. *Fan Motor Burned*, *Pipe Leakage Behind Wall*).

5. **Operational Priority & Time Window:**
   * **Priority Level:**
     - **Normal:** Standard routine repair (SLA: 24 - 48 Hours).
     - **Emergency:** Life safety, water flooding, power outage, or gas leak (SLA: 2 Hours).
     - **Availability Time:** Inquire occupant's preferred inspection window (e.g. `09:00 - 12:00` or `14:00 - 17:00`).
6. **Detailed Remarks & Photo Attachments:**
   * Enter precise diagnostic notes provided by the occupant.
   * Attach photo evidence if uploaded by the resident or captured on-site.
7. **Submission:** Click **"Submit Ticket"**. The system generates a unique **Tracking ID** and sets status to **Unassigned (2)**. An automated SMS alert is dispatched to the occupant.

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4. Engineering Dispatch & Field Supervision (GE Office & Field Supervisor Guide)

The **Garrison Engineer (GE)** and Maintenance Supervisors govern technician assignments, job tracking, and field exceptions.

4.1 Ticket Dispatching Workflow

1. Navigate to **Complaints Board** (`/admin/complaints`).
2. Filter the grid by **Status: Unassigned** and your assigned **Sector**.
3. Click the target ticket to inspect complaint parameters and occupant availability.
4. In the **Assignment Section**:
   * Open the **Assigned Employee** dropdown. The system indicates active job counts beside each technician's name to prevent staff overloading.
   * Select a qualified technician matching the trade category.
5. Click **"Assign Technician"**. The ticket status transitions to **Assigned (3)**.

4.2 Job Dispatch Slip Printing

- On the complaint details view, click **"Print Dispatch Slip"** (`/admin/complaints/{id}/print-slip`).
- The system produces a physical job card containing:
- Official Navy CMS header and QR code.
- House number, occupant rank/name, and contact number.
- Fault description and reported symptoms.
- Assigned technician badge details.
- Blank signature block for resident sign-off upon completion.

4.3 Field Exception Workflows

When technicians arrive on-site, unexpected circumstances may prevent immediate resolution. Technicians report back to the maintenance supervisor to apply specific status codes:

Field Condition	System Action	Status Code	Required Operational Action
House Locked / Occupant Absent	Mark as **"Door Lock"**	`11`	Automated SMS notification sent to resident. Rescheduled for subsequent availability window.
Material / Part Not in Store	Mark as **"Product N/A"**	`8`	Triggers procurement alert to Storekeeper to initiate replenishment purchase order.
Unauthorized Structural Alteration	Mark as **"Un-Authorized"**	`9`	Work suspended immediately. Incident flagged to GE / CMES for official structural inquiry.
Damage due to Resident Negligence	Mark as **"Barrack Damages"**	`10`	Form B drafted for cost recovery. Repair deferred until damage assessment approval.
Routine Fix Complete	Mark as **"Resolved"**	`1`	Ticket marked resolved pending resident confirmation and star rating.
Final Closure	Mark as **"Closed"**	`4`	Administrative closure after inspection and quality verification.

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5. Material Requisition & The Performa Approval Engine

When maintenance requires parts beyond standard service toolkits, formal financial and inventory sanctions must be recorded.

5.1 Performa Classification Hierarchy

CMS enforces three distinct approval levels depending on material valuation:

- **Work Performa (Code 4):** Minor consumables (e.g. tap washers, light switches, fuses). Sanctioned directly at GE level.
- **Maintenance Performa (Code 5):** Standard structural replacements (e.g. sanitary fixtures, ceiling fan rewinding, distribution boards). Sanctioned by the **Garrison Engineer (GE)**.
- **Priced Performa (Codes 6 & 7):** Major capital replacements exceeding monetary threshold (e.g. main distribution cable, water pump motor replacement). Mandates formal digital signature by **Executive Engineer (XEN)** or **CMES** in concurrence with the **GE**.

5.2 Creating a Material Performa (Field Supervisor / In-Charge)

1. Open the active complaint and scroll to **Material Requisition** section.
2. Click **"Generate Material Performa"**.
3. Select appropriate **Performa Type** (Work, Maintenance, or Priced).
4. Click **"Add Spare Part"**:
   * Select item from the verified warehouse catalog.
   * System displays real-time warehouse availability.
   * Input required **Quantity** and justification remarks.
5. Repeat for all required items.
6. Click **"Submit for Approval"**. The Performa voucher transitions to `Pending` status.

5.3 Authorizing Performas (GE / Garrison Engineer Approval Authority)

1. Navigate to **Approvals** → **Pending Performas** (`/admin/approvals`).
2. Inspect the voucher line items, estimated financial cost, and complaint history.
3. The **Garrison Engineer (GE)** or designated sanctioning officer reviews the request:
   * **Approve:** Performa status changes to `Approved` . System reserves stock in warehouse and generates a store issuance token.
   * **Reject:** Provide mandatory rejection rationale. Ticket returns to the maintenance supervisor for alternate engineering solution.
   * **Revise:** Modify sanctioned quantities or substitute specified items before granting approval.

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6. Warehouse & Spares Inventory Management (Storekeeper Guide)

The Storekeeper manages warehouse inventory balances, inward stock deliveries, and outward dispatches against sanctioned vouchers.

6.1 Inward Stock Entry (Receiving Goods)

1. Navigate to **Inventory** → **Spares Catalog** (`/admin/spares`).
2. Click **"Add Stock"** (`/admin/spares/{id}/add-stock`).
3. Enter inward delivery parameters:
   * **Vendor / Contractor:** Select certified supplier.
   * **Purchase Order (PO) / Delivery Challan:** Reference number.
   * **Batch / Lot Number:** If applicable for warranty tracking.
   * **Quantity Received:** Numerical count.
   * **Unit Cost:** Cost per unit in PKR.
4. Click **"Save Inward Stock"**. The ledger balance updates instantaneously, and an immutable record is registered in `stock_add_data` .

6.2 Outward Material Issuance against Sanctioned Performa

1. Open **Approvals** → **Approved Performas** or **Spares** → **Issue Stock**.
2. Scan or enter the sanctioned **Performa Token Number**.
3. Verify that the collecting technician's identity matches the assigned work order.
4. Click **"Issue Stock"**.
5. **System Effects:**
   * Sanctioned quantities are decremented from `spares.current_stock` .
   * Items are permanently debited to the complaint's expenditure ledger in `complaint_spares` .
   * Click **"Print Material Issuance Slip"** (`/admin/spares/{id}/print-slip`) for store archives.

6.3 Low Stock & Reorder Thresholds

The catalog dashboard incorporates visual threshold indicators:
* **Green (Normal):** Current stock is safely above safety buffer (`current_stock > minimum_stock`).
* **Amber (Warning):** Stock has reached or dropped below reorder level (`current_stock ⩽ minimum_stock`). Storekeeper must prepare procurement requisition.
* **Red (Depleted):** Zero inventory balance. System automatically restricts dispatching this part on new performas.

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7. Service Level Agreements (SLA) & Quality Assurance

7.1 SLA Architecture

CMS monitors ticket progression against trade-specific time benchmarks:

Category	Priority	Max Response Time (Dispatch)	Max Resolution Time	Escalation Threshold
Gas Leakage / Fire Hazard	Critical	15 Minutes	2 Hours	30 Minutes
Main Water Pipe Burst	Emergency	30 Minutes	4 Hours	1 Hour
Total Electrical Blackout	Emergency	30 Minutes	4 Hours	1 Hour
Routine Electrical / Plumbing	Normal	4 Hours	24 Hours	12 Hours
Civil / Carpentry / Painting	Normal	8 Hours	48 Hours	24 Hours

7.2 Resident Feedback & Rating Engine

1. Following resolution, the resident is prompted via mobile app or SMS link to review the service.
2. Resident rates service on a **1 to 5 Star Scale** and leaves qualitative remarks.
3. **Automated Escalation Rule:**
   * Any rating of **1 or 2 Stars** triggers an automated quality audit alert on the GE and CMES dashboards.
   * The GE / Maintenance Supervisor must contact the occupant within 24 hours to address grievances and record corrective actions.

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8. Master Data & Administrative Governance

System Administrators maintain structural directories, user credentials, and platform governance.

8.1 Staff & Technician Onboarding

1. Navigate to **Employee Management** (`/admin/employees`).
2. Click **"Add New Employee"**.
3. Input official particulars: Full Name, CNIC, Official Designation, Trade Category (e.g. Lineman, Plumber, Mason), Mobile Number, and Base Station.
4. Set status to **Active (1)**.

8.2 House & Accommodation Directory (Batch CSV Operations)

1. Navigate to **House Management** (`/admin/houses`).

2. To import batch accommodation records:
   * Click **"Download Sample CSV"** (`/admin/houses/sample-csv`).
   * Populate required columns: `city_id` , `cme_id` , `sector_id` , `house_no` , `occupant_rank` , `occupant_name` , `mobile_no` .
   * Click **"Import Houses"** and upload the verified CSV spreadsheet.
3. System validates records and indexes accommodations for helpdesk autocomplete.

8.3 Mobile Device Binding Governance (`/admin/registered-devices`)

To ensure operational security, resident accounts are cryptographically bound to authorized mobile hardware:
* Each house account may register a maximum of authorized handheld devices.
* **Device Reset Procedure:** When an officer or resident changes mobile handsets and receives a *"Device Registration Mismatch"* warning:
1. Open **Registered Devices** (`/admin/registered-devices`).
2. Search by House Number or Officer Rank.
3. Verify resident identity through official naval directory.
4. Click **"Revoke / Reset Device Binding"**.
5. The resident can immediately log in from the new device, establishing a clean security bond.

8.4 Role-Based Access Control (RBAC) Management

- CMS implements strict **Exact-Match Permission Verification** (`in_array($module, $permissions, true)`).
- To configure role privileges:
  1. Open **Role Management** (`/admin/roles`).
  2. Select target role (e.g. *Helpdesk Operator*, *Storekeeper*).
  3. Explicitly tick individual module permission checkboxes.
  4. Save. Changes take effect on next user login session.

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9. Reports, Analytics & Executive Decision Support

9.1 Standard Reporting Suites

Navigate to **Reports & Analytics** (`/admin/reports`) to extract verified audit documentation:

- **Complaint Volume & Resolution Report:** Aggregates total logged tickets, active jobs, resolved counts, and average resolution times broken down by City, Sector, and Trade.
- **Technician Performance Matrix:** Evaluates individual technician output, average turnaround time, active backlog, and resident satisfaction score.
- **Spares Consumption & Valuation Audit:** Itemized breakdown of materials issued per accommodation and category, comparing budget allocations against actual consumption.
- **SLA Breach & Escalation Report:** Logs all tickets that exceeded prescribed response or resolution windows with recorded supervisor justifications.

9.2 Export Protocols

All analytical datasets can be generated in three standardized formats:
1. **Interactive Screen View:** High-level executive charts with drill-down capability.
2. **Executive PDF Memo:** Pre-formatted, publication-ready A4 documentation containing official naval headers, sign-

off blocks, and summary statistics.
3. **Raw CSV / Excel Data:** Tabular dumps for external auditing, spreadsheet analysis, and financial reconciliation.

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10. Operational Troubleshooting & Frequently Asked Questions (FAQ)

10.1 Access & Authentication Issues

- **Q: A staff member receives a "403 Forbidden - Unauthorized Action" error.**
- *Resolution:* Navigate to **Role Management** (`/admin/roles`), inspect the assigned role, and ensure the specific sub-permission (e.g. `complaints.view` , `spares.add-stock`) is explicitly checked. Have the user log out and log back in.
- **Q: A technician does not appear in the assignment dropdown list.**
- *Resolution:* Ensure the employee account is set to `status = 1` (Active) and their designated trade category matches the complaint discipline.

10.2 Workflow & Ticket Issues

- **Q: Can a complaint be marked "Resolved" while a Material Performa is still pending?**
- *Resolution:* No. The system state machine prevents ticket closure if an associated performa is in `pending` or `draft` state. The performa must either be formally approved or cancelled first.
- **Q: How to handle duplicate complaints lodged by multiple family members?**
- *Resolution:* Open the duplicate ticket, navigate to status update, and select **"Closed"** with remarks: *"Duplicate of Ticket #[Original_ID]"*.

10.3 Inventory & Stock Discrepancies

- **Q: The system shows zero stock for an item that is physically present in the warehouse.**
- *Resolution:* The Storekeeper must perform an **Inward Stock Entry** referencing the delivery voucher before the system will authorize warehouse dispatch.
